

Vehicle Dynamics Lap-Time Simulator

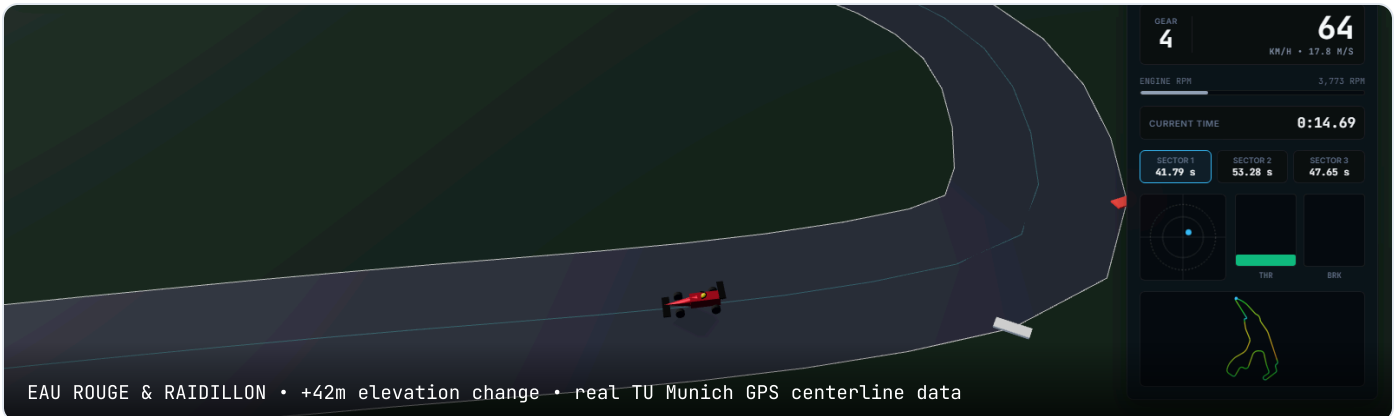
A from-scratch lap-time solver — friction circle, two-pass kinematics, Pacejka tire model — driving a live 3D replay of Spa-Francorchamps.

MATLAB

THREE.JS / WEBGL

PACEJKA '94

VANILLA JS PHYSICS



3:19.18

LAP TIME

180.7 km/h

TOP SPEED

1.72 g

PEAK LATERAL G

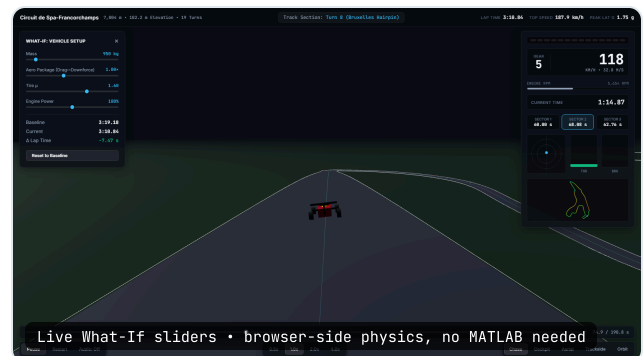
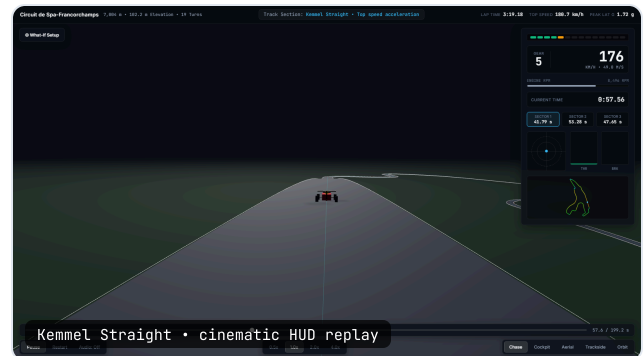
1,401

TRACK WAYPOINTS

CORNERING SPEED LIMIT

$$v_{\text{corner}} = \sqrt{\frac{\mu g}{\kappa - K_{\text{aero}}}}$$

Max cornering speed balances tire grip (μg) against track curvature (κ), reduced by aero downforce grip gain (K_{aero}). A friction-circle solver then runs a forward/backward two-pass integration to combine this with engine power and braking limits into a full speed profile.



PHYSICS ENGINE

- **Friction-circle** traction model combining lateral grip and longitudinal accel/brake force
- **Two-pass kinematic solver** (forward acceleration + backward braking) iterated to convergence
- **Pacejka '94 Magic Formula** nonlinear tire model with load-sensitive grip, alongside a classic linear Coulomb fallback
- Aero downforce/drag, gearbox torque-curve powertrain, real track curvature from GPS centerline data

STACK & DELIVERY

- **MATLAB** source-of-truth solver, test suite, and native 3D animation (App Designer)
- **Three.js / WebGL** cinematic browser replay with F1-style HUD, shift lights, live minimap
- Physics engine **ported to vanilla JS** so the What-If panel re-solves the lap client-side in ~10ms — no MATLAB license required to try it
- Open track data: TU Munich racetrack database (Spa-Francorchamps, CC BY 4.0)

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github.com/RoanMerluccio/vehicle-dynamics-laptime-sim

